

arduino_code.cpp

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1  /*
2  Task: Decimal to Binary Conversion Using LEDs in Tinkercad
3
4  Using Arduino in Tinkercad, design and implement a circuit that converts a decimal number entered
   through the Serial Monitor into its corresponding 4-bit binary representation.
5
6  What is 4-bit Binary?
7
8  A binary number uses only two digits: 0 and 1.
9
10 In a 4-bit binary number, there are 4 positions. Each position represents a power of 2:
11
12 8 4 2 1
13 ↓ ↓ ↓ ↓
14 23 22 21 20
15
16 For example:
17
18 10 = 8 + 2
19
20 Therefore:
21
22 10 = 1010
23
24 Here, the leftmost bit represents 8 and the rightmost bit represents 1.
25
26 MSB and LSB
27
28 MSB means Most Significant Bit. It is the leftmost bit of a binary number and has the highest
   value.
29
30 LSB means Least Significant Bit. It is the rightmost bit and has the lowest value.
31
32 For example, in:
33
34 1010
35
36 The positions are:
37
38 1 0 1 0
39 ↑ ↑
40 MSB LSB
41
42 For this task, LED 1 should represent the MSB and LED 4 should represent the LSB.
43
44 So for 10 (1010):
45
46 LED 1 → ON → 1
```

```
47 LED 2 → OFF → 0
48 LED 3 → ON → 1
49 LED 4 → OFF → 0
50
51 Requirements
52
53 1. Create the circuit in Tinkercad Circuits using:
54 Arduino Uno
55 4 LEDs
56 4 appropriate resistors
57 Breadboard and jumper wires
58 2. Use the Serial Monitor to take a decimal number as input from the user.
59
60 3. Accept decimal numbers from 0 to 15.
61
62 4. Convert the entered decimal number into its 4-bit binary representation using your Arduino
63 program.
64
65 5. Do not use a ready-made decimal-to-binary conversion function. Perform the binary conversion
66 yourself in the program.
67
68 6. Display the binary representation using the four LEDs:
69 LED ON → 1
70 LED OFF → 0
71
72 7. The LEDs must represent the bits from MSB to LSB.
73
74 Example
75
76 If the user enters:
77
78 10
79
80 The Arduino should convert it to:
81
82 1010
83
84 The LEDs should be:
85
86 LED 1 LED 2 LED 3 LED 4
87 ON OFF ON OFF
88
89 Make sure the circuit works correctly for different inputs from 0 to 15.
90
91 Code by Rayan
92 */
93
94 int led_1 = 2;
95 int led_2 = 3;
96 int led_3 = 4;
```

```
95  int led_4 = 5;
96
97  void setup()
98  {
99      pinMode(led_1, OUTPUT);
100     pinMode(led_2, OUTPUT);
101     pinMode(led_3, OUTPUT);
102     pinMode(led_4, OUTPUT);
103     Serial.begin(9600);
104     Serial.print("Enter  a decimal number between 0 and 15: ");
105 }
106
107 void loop()
108 {
109     if(Serial.available()>0){
110         int dec_num = Serial.parseInt();
111         Serial.println(dec_num);
112         while (Serial.available() > 0) {
113             Serial.read();
114         }
115         int led_arr[4]={0,0,0,0};
116         if(dec_num>=0 && dec_num<=15){
117             for(int i = 3 ; i >=0; i--){
118                 led_arr[i] = dec_num%2;
119                 dec_num /=2;
120             }
121             Serial.print("The binary number is:");
122             for (int i = 0; i < 4; i++) {
123                 Serial.print(led_arr[i]);
124             }
125             Serial.println();
126             digitalWrite(led_1,led_arr[0]);
127             digitalWrite(led_2,led_arr[1]);
128             digitalWrite(led_3,led_arr[2]);
129             digitalWrite(led_4,led_arr[3]);
130             delay(2000);
131             digitalWrite(led_1,LOW);
132             digitalWrite(led_2,LOW);
133             digitalWrite(led_3,LOW);
134             digitalWrite(led_4,LOW);
135             Serial.println("Display done");
136             Serial.print("Enter  a decimal number between 0 and 15: ");
137         }else{
138             Serial.println("Enter in valid range....");
139             Serial.print("Enter  a decimal number between 0 and 15: ");
140         }
141     }
142 }
```